

12.263

EE25BTECH11042 - Nipun Dasari

Question:

The matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 2 \end{pmatrix} \quad (0.1)$$

can be decomposed into the product of a lower triangular matrix $\mathbf{L}$ and an upper triangular matrix $\mathbf{U}$ as $\mathbf{A} = \mathbf{LU}$. Let $\mathbf{x}, \mathbf{z} \in \mathbb{R}^3$ and $\mathbf{b} = (1 \ 1 \ 1)^\top$.

- Find the solution $\mathbf{z}$ of the system $\mathbf{Lz} = \mathbf{b}$.
- Find the solution $\mathbf{x}$ of the system $\mathbf{Ux} = \mathbf{z}$.

Solution:

LU Decomposition by Row Transformations

We will perform row operations on $\mathbf{A}$ to obtain $\mathbf{U}$. The multipliers used will form the entries of $\mathbf{L}$.

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 2 \end{pmatrix} \quad (2.1)$$

Operation $R_2 \rightarrow R_2 - (2)R_1$. The multiplier is $l_{21} = 2$.

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 1 & 3 & 2 \end{pmatrix} \quad (2.2)$$

Operation $R_3 \rightarrow R_3 - (1)R_1$. The multiplier is $l_{31} = 1$.

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 2 & 1 \end{pmatrix} \quad (2.3)$$

Operation $R_3 \rightarrow R_3 - (-2)R_2$. The multiplier is $l_{32} = -2$.

$$\mathbf{U} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (2.4)$$

The lower triangular matrix $\mathbf{L}$ is constructed from the multipliers.

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -2 & 1 \end{pmatrix} \quad (2.5)$$

Part a): Solving $Lz = b$

We solve the system using Gaussian elimination on the augmented matrix $[L|b]$.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 2 & 1 & 0 & 1 \\ 1 & -2 & 1 & 1 \end{array}\right) \begin{array}{c} \xleftarrow{R_2 \rightarrow R_2 - 2R_1} \\ \xleftarrow{R_3 \rightarrow R_3 - R_1} \end{array} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -2 & 1 & 0 \end{array}\right) \quad (2.6)$$

$$\xleftarrow{R_3 \rightarrow R_3 + 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -2 \end{array}\right) \quad (2.7)$$

From the reduced row echelon form, we get the solution:

$$\mathbf{z} = \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \quad (2.8)$$

Part b): Solving $Ux = z$

We solve the system using the augmented matrix $[U|z]$.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & -2 \end{array}\right) \xleftarrow{R_2 \rightarrow -R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right) \quad (2.9)$$

$$\begin{array}{c} \xleftarrow{R_1 \rightarrow R_1 - R_3} \\ \xleftarrow{R_1 \rightarrow R_1 - R_2} \end{array} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right) \quad (2.10)$$

From the reduced row echelon form, we get the solution:

$$\mathbf{x} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \quad (2.11)$$